A Bayesian Varying Coefficient Multiple Index Model (BVCMIM) for Longitudinal Analysis of Environmental Chemical Mixtures

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1 Abstract

This paper introduces the Bayesian Varying Coefficient Multiple Index Model (BVCMIM), a novel statistical approach designed to estimate the longitudinal effects of environmental chemical exposure mixtures on human health outcomes. Traditional methods in environmental health often focus on single chemical exposures or rely on linear models that may not capture the complex, non-linear, and non-additive relationships inherent in chemical mixtures. BVCMIM overcomes these limitations by allowing for the estimation of non-linear relationships between exposure indices and health outcomes, while also accounting for

interactions among different chemical exposures over time. The model incorporates the use of continuous shrinkage priors for sparsity, ensuring that only the most relevant exposures are included in the analysis, and applies Hamiltonian Monte Carlo for uncertainty quantification. Through a series of simulations, we demonstrate that BVCMIM provides robust and interpretable estimates of both baseline and longitudinal health effects, even in scenarios with intricate chemical interactions. BVCMIM offers a flexible and powerful tool for environmental health researchers aiming to understand the impact of complex chemical mixtures on health trajectories.

2 Introduction

Exposure to chemical stressors can have diverse effects on human health [28, 24, 11]. Historically, research has primarily focused on modeling the impact of exposure to individual chemicals [13, 10]. However, in reality, environmental exposures typically consist of complex mixtures of chemicals, making it critical to understand their combined effects. This is a key research priority for the National Institute of Environmental Health Sciences (NIEHS) [6, 25]. When analyzing mixture effects, researchers often aim to assess how exposures influence health outcomes across different ages, necessitating a longitudinal approach [5]. Longitudinal designs allow researchers to capture time-varying exposure patterns and health responses, providing insights into both transient and cumulative effects of chemical mixtures [12]. Such designs are essential for identifying critical windows of susceptibility, particularly during prenatal and early childhood periods when biological systems are rapidly developing and highly sensitive to environmental stressors [22]. Moreover, repeated measures of exposures and outcomes enhance causal inference and statistical efficiency by accounting for within-subject variability and controlling for time-invariant confounders [9]. By modeling exposure trajectories and cumulative burden across developmental stages, longitudinal mixture analysis provides a more comprehensive understanding of how complex environmental exposures jointly influence health across the life course.

Several statistical approaches have been proposed to analyze mixture effects, including exposure-response surface methods and exposure-index methods. Exposure-response surface approaches, like Bayesian kernel machine regression (BKMR) [3], estimate multi-dimensional relationships between exposures and outcomes in a non-parametric way, capturing linear and non-linear effects, and non-additive interactions effects. For BKMR, interpretation relies mostly on examining the posterior analysis of the high-dimensional exposure-response surface, heavily utilizing visualization. When dealing with a moderate to large number of components in the mixture, rigorous interpretation becomes challenging due to the vast number of main effects and interactions involved. For instance, in the analysis of 18 exposures discussed in this study, BKMR necessitates visually inspecting 18 main effect exposure-response functions and 306 pairwise interaction surface plots. Furthermore, BKMR struggles with high-

dimensional data, a phenomenon known as the "curse of dimensionality" [27].

On the other hand, exposure-index methods assess the association between an outcome and a composite index formed from multiple exposures. Two widely used approaches in environmental health are Weighted Quantile Sum (WQS) regression [7] and Quantile G-computation (QGC) [18]. Both methods begin by transforming the exposure variables into quantiles and then fitting a linear model to estimate the effect of the combined exposures. In WQS regression, the exposure index is a weighted sum of quantile transformed exposures, where the weights are estimated from the data and constrained to be non-negative and sum to one. These weights reflect the relative contribution of each exposure to the overall mixture effect. The model then estimates a single coefficient—the overall index effect—that quantifies the association between the composite exposure and the outcome. Quantile G-computation generalizes this idea by relaxing the constraint on weights and allowing them to be negative or positive, thereby enabling estimation of both harmful and protective components within the mixture. A key advantage of these methods is interpretability: they provide a single parameter representing the mixture's overall effect, along with component-specific weights that indicate the relative influence of each exposure. This simplification leads to clearer communication of results. Additionally, their compact structure enables efficient inference when model assumptions are valid. However, because these methods rely on linear modeling frameworks, they may lack the flexibility needed to capture complex, nonlinear, or interactive exposure–response relationships. While it is possible to modify these methods to include higher-order terms or interactions, doing so parametrically often complicates the interpretability of the weights and can lead to inaccurate inference.

Recently, the Bayesian Multiple Index Model (BMIM) was introduced [21] as a flexible framework that combines features of both exposure–response surface methods and exposure-index models for non-longitudinal data. BMIM assumes that chemicals can be grouped naturally based on shared characteristics or similar modes of action, and it constructs a separate index for each group. This multiple-index approach reduces the dimensionality of the exposure space while allowing for non-linear relationships between each index and the outcome, as well as non-additive interactions among exposures. As a result, BMIM offers improved flexibility for modeling complex mixture effects.

In longitudinal settings—where exposures are measured once but health outcomes are recorded at multiple time points—it becomes important to evaluate not only the effect of the exposure mixture on a single outcome, but also on the trajectory of the outcome over time. To address this, previous approaches have integrated exposure–response or exposure-index models with mixed-effects frameworks to capture both baseline levels and changes in health outcomes. For instance, Bayesian Varying Coefficient Kernel Machine Regression (BVCKMR) [20] combines Bayesian kernel machine regression with mixed-effects modeling to allow for nonlinear exposure effects that vary over time. Similarly, BVCQR [15] extends quantile regression to a mixed-effects setting, relaxing the directional constraint used in WQS to allow more flexibility in estimating mixture effects. These methods enable estimation of mixture effects on both the baseline out-

come and its trajectory across time. However, they also inherit the limitations of the underlying exposure–response frameworks on which they are built.

In summary, despite recent advancements in statistical methods for analyzing chemical mixtures, substantial challenges remain—particularly in longitudinal settings. Many existing models, such as Bayesian Kernel Machine Regression (BKMR), offer flexible surface estimation but suffer from scalability issues and limited interpretability when applied to high-dimensional exposures. Index-based approaches like Weighted Quantile Sum (WQS) regression improve interpretability by summarizing mixtures, but are typically linear, cross-sectional, and rely on restrictive assumptions (e.g., directional constraints, quantile transformation). Moreover, these models often cannot capture the evolution of exposure effects over time. In longitudinal studies of child health, understanding how mixture effects change with age is crucial for identifying sensitive windows of development.

To address these limitations, we propose a Bayesian Varying Coefficient Multiple Index Model (BVCMIM) that (i) incorporates group-wise exposure indices to enhance interpretability, (ii) flexibly models nonlinear effects, (iii) allows for exposure group interactions using kernel functions, and (iv) integrates subject-specific time-varying random effects for repeated outcome measurements. Therefore, BVCMIM bridges the gap between interpretability, flexibility, and scalability—providing a practical framework for analyzing complex environmental mixtures in longitudinal cohort studies.

3 Data Overview and Preparation

To illustrate the applicability of the proposed method described in later sections, we analyze data from the Health Outcomes and Measures of the Environment (HOME) Study—a prospective birth cohort conducted in the Greater Cincinnati, Ohio metropolitan area. The HOME Study enrolled pregnant women from three local hospitals and collected extensive environmental, behavioral, and developmental data on their children at multiple time points.

For this study, we focus on assessing the relationship between prenatal chemical exposures and children's neurodevelopment, as measured by the Bayley Mental Development Index (MDI) [1]. MDI scores were assessed at child ages 12, 24, and 36 months. Chemical exposures data were collected during pregnancy from maternal biological samples. Our analytic subset includes 149 children with complete data on relevant exposures and neurodevelopmental outcomes, consistent with previous work [15]. The demographics information for all participants are provided in table 1.

Following the preprocessing steps outlined in Braun et al. [4], we excluded chemicals with detection frequencies below 20%, yielding 36 continuous chemical exposure variables that can be grouped naturally into five sets of pollutants hypothesized to act similarly: Phthalates and Bisphenol A (Pth & BPA), Polychlorinated Biphenyls (PCB), Organochlorines (OC), Polybrominated Diphenyl Ethers (PBDE) and Perfluorinated Compounds (PFC). For measurements below

the limit of detection (LOD), we imputed values using the standard substitution method: LOD/$\sqrt{2}$ [14]. The geometric means and standard deviations for 36 chemical exposure variables are provided in table 2. To mitigate the influence of skewed distributions and potential outliers, all chemical concentrations were log-transformed using $\log_{10}$.

This prepared dataset provides a rich platform to evaluate the proposed model in the context of high-dimensional, longitudinal environmental health data.

4 Methodological Background

4.1 Bayesian Multiple Index Model (BMIM)

Suppose there are P chemical exposures denoted by $z_{i1}, \ldots, z_{iP}$ for subject $i = 1, \ldots, n$, which can be naturally partitioned into M mutually exclusive groups. Let $\boldsymbol{z}_{im} = (z_{im1}, \ldots, z_{imL_m})^T$ represent the vector of exposures in group $m = 1, \ldots, M$, where L_m is the number of chemicals in group m. The Bayesian Multiple Index Model (BMIM) models the outcome y_i as:

$$y_i = h(\boldsymbol{z}_{i1}^T\boldsymbol{\theta}_1, \ldots, \boldsymbol{z}_{iM}^T\boldsymbol{\theta}_M) + \boldsymbol{x}_i^T\boldsymbol{\beta} + \epsilon_i \tag{1}$$

Here, $\boldsymbol{\theta}_m$ is a vector of index weights for chemicals in group m, and the function $h(\cdot)$ is a flexible surface (e.g., via kernel regression) that maps the M group-specific indices to the response. $\boldsymbol{x}_i$ is a vector of additional covariates with associated coefficients $\boldsymbol{\beta}$, and $\epsilon_i \sim N(0, \sigma^2)$ is the residual error term.

4.2 Mixed-Effects Modeling for Longitudinal Data

A mixed-effects model is an extension of the standard linear regression model that is effective for analyzing and modeling data generated from multiple sources of variation. It establishes a relationship between a response variable (outcome) and several explanatory variables. What distinguishes the mixed-effects model is its incorporation of both fixed and random factors. Fixed effects estimate average responses across groups, similar to a standard regression model, while random effects (such as individual-specific effects) account for natural variability in responses among different entities, facilitating response estimation for each entity in the analysis. It is essential to consider the correlation among measurements within the same entity during modeling, as these measurements are often correlated. The correlation among repeated responses can vary, resulting in different covariance or correlation structures. Under a longitudinal study, each subject has the same outcome measured at different time points, or more specifically, ages. Measures for the same subject are correlated, so, random effects can be add for each subject to account for inter-subject variations. The model accommodates various covariance structures and enables the estimation of effects and variance parameters. The dataset's structure can vary from bal-

anced (equal observations per entity) to unbalanced (unequal observations per entity), and the model can handle such variations effectively.

5 Proposed Method: Bayesian Varying Coefficient Multiple Index Model (BVCMIM)

5.1 Model Specification

Let y_{ij} denote the health outcome for subject i at age age_{ij}. Let $z_{i1}, \ldots, z_{iP}$ be the P chemical exposures for subject i, which can be partitioned into M mutually exclusive groups. Define $\boldsymbol{z}_{im} = (z_{im_1}, \ldots, z_{im_{L_m}})^T$ as the vector of exposures in group $m = 1, \ldots, M$, where L_m is the number of chemicals in group m. Fowlling the mixed effect model specification [20], the proposed Bayesian Varying Coefficient Multiple Index Model (BVCMIM) is formulated as:

$$\begin{aligned} y_{ij} =& h_1^M(\boldsymbol{z}_{i1}^T\boldsymbol{\theta}_{11}, \ldots, \boldsymbol{z}_{iM}^T\boldsymbol{\theta}_{1M}) + h_2^M(\boldsymbol{z}_{i1}^T\boldsymbol{\theta}_{21}, \ldots, \boldsymbol{z}_{iM}^T\boldsymbol{\theta}_{2M}) \cdot age_{ij} \\ &+ \boldsymbol{x}_i^T\boldsymbol{\beta} + \boldsymbol{u}_{ij}^T\boldsymbol{b}_i + \gamma_1 + \gamma_2 \cdot age_{ij} + \epsilon_{ij} \end{aligned} \tag{2}$$

Here, $h_q^M(\cdot)$, for $q = 1, 2$, are flexible functions—parametric or nonparametric—that model the combined effects of M exposure indices. The vector $\boldsymbol{\theta}_{qm}$ contains weights of length L_m that quantify the contribution of each chemical in group m to the q-th index. Specifically, $h_1^M(\cdot)$ estimates the mixture effects on the outcome at baseline age ($age_{ij} = 0$), while $h_2^M(\cdot)$ captures how these effects evolve over time. The terms γ_1 and γ_2 represent fixed intercept and slope, respectively. $\boldsymbol{x}_i$ denotes additional covariates with coefficients $\boldsymbol{\beta}$, $\boldsymbol{u}_{ij}^T\boldsymbol{b}_i$ accounts for subject-specific random effects, and $\epsilon_{ij} \sim N(0, \sigma^2)$ is the residual error.

This model can be further written in matrix form as

$$\boldsymbol{Y_i} = \boldsymbol{X}_i\boldsymbol{\beta} + \boldsymbol{W}_i\boldsymbol{h}_i + \boldsymbol{U}_i\boldsymbol{b}_i + \boldsymbol{\epsilon}_i \tag{3}$$

for $i = 1, \ldots, n$ and repeated measures $j = 1, \ldots, J_i$ and $\boldsymbol{b}_i \sim N_2(0, \sigma^2\boldsymbol{D})$ and $\boldsymbol{\epsilon}_i \sim (0, \sigma^2\boldsymbol{I}_{J_i})$. $\boldsymbol{h}_i$ is the vector of mixture effects and $\boldsymbol{b}_i$ is the vector of random effects for subject i. $\boldsymbol{U}$ and $\boldsymbol{W}$ are matrices constructed by ages of different individuals. X_i is the matrix of covariates for subject i with coefficients $\boldsymbol{\beta}$. $\boldsymbol{Y}_i = (Y_{i1}, \ldots, Y_{iJ_i})^T$ is the outcome for subject i. The fixed effects γ_1 and γ_2 can be included by adding column of intercept and age into X_i Specifically, we write the model as following:

$$\boldsymbol{Y} = \begin{pmatrix} \boldsymbol{Y}_1 \\ \boldsymbol{Y}_2 \\ \vdots \\ \boldsymbol{Y}_n \end{pmatrix} = \begin{pmatrix} Y_{11} \\ \vdots \\ Y_{1J_1} \\ \vdots \\ Y_{n1} \\ \vdots \\ Y_{nJ_n} \end{pmatrix}, \boldsymbol{\beta} = \begin{pmatrix} \beta_1 \\ \vdots \\ \beta_n \end{pmatrix}, \boldsymbol{X} = \begin{pmatrix} \boldsymbol{X}_1 \\ \vdots \\ \boldsymbol{X}_n \end{pmatrix}$$

$$\boldsymbol{U} = \begin{pmatrix} 1 & age_{11} & & & & & & \\ \vdots & \vdots & & & & & & \\ & & 1 & age_{21} & & & & \\ & & \vdots & \vdots & & & & \\ & & 1 & age_{2J_1} & & & & \\ & & & & \ddots & & & \\ & & & & & 1 & age_{n1} & \\ & & & & & \vdots & \vdots & \\ & & & & & 1 & age_{nJ_n} & \end{pmatrix}$$

$$\boldsymbol{b} = \begin{pmatrix} b_{1,1} \\ b_{2,1} \\ b_{1,2} \\ b_{2,2} \\ \vdots \\ \vdots \\ b_{1,n} \\ b_{2,n}, \end{pmatrix}, \boldsymbol{W} = \begin{pmatrix} 1 & 0 & \dots & 0 & age_{11} & 0 & \dots & \\ \vdots & & & & \vdots & & & \\ 1 & 0 & \dots & 0 & age_{1J_1} & 0 & \dots & \\ 0 & 1 & 0 & \dots & 0 & age_{21} & 0 & \dots \\ 0 & \vdots & & & & \vdots & & \\ 0 & 1 & 0 & \dots & 0 & age_{2J_2} & 0 & \dots \\ 0 & 0 & 1 & 0 & \dots & 0 & age_{31} & 0 \\ 0 & 0 & \vdots & & & & \vdots & \ddots \end{pmatrix},$$

$$\boldsymbol{h} = \begin{pmatrix} h_{1,1} \\ h_{1,2} \\ \vdots \\ h_{1,n} \\ h_{2,1} \\ h_{2,2} \\ \vdots \\ h_{2,n} \end{pmatrix}$$

5.2 Prior Distributions and Posterior Inference

Following the Bayesian parameterization of the mixed effects model [20], we can express our proposed model as a hierarchical model:

$$\boldsymbol{y}|\boldsymbol{h}, \boldsymbol{X}, \boldsymbol{\beta}, \sigma^2, \boldsymbol{b} \sim N(\boldsymbol{X\beta} + \boldsymbol{Wh} + \boldsymbol{Ub}, \sigma^2\boldsymbol{I}) \tag{4}$$

$$\boldsymbol{h}|\boldsymbol{\theta}_1, \boldsymbol{\theta_2} \sim N_{2n}\left(\boldsymbol{0}, \begin{pmatrix} \boldsymbol{K}_1 & 0 \\ 0 & \boldsymbol{K}_2 \end{pmatrix}\right) \tag{5}$$

where $\boldsymbol{K}_q$ denotes the kernel matrix with i, j element $K_q(\boldsymbol{z}^*_{qi}, \boldsymbol{z}^*_{qj})$. Depending on the problem, several different kernel functions K_q can be employed. To capture non-linear relationships between the exposure indices and the outcome, we model the index-response functions $h_1^M(\cdot)$ and $h_2^M(\cdot)$ using Gaussian kernel functions. Specifically, for each function $h_q^M(\cdot)$, where $q = 1, 2$, we define a kernel matrix $K_q \in \mathbb{R}^{n\times n}$ with elements:

$$K_q(z_i^*, z_j^*) = \eta_q \exp\left(-\sum_{m=1}^{M} \rho_{qm}(z_{im}^* - z_{jm}^*)^2\right),$$

where $z_i^* = (z_{i1}^T\boldsymbol{\theta}_{q1}, \ldots, z_{iM}^T\boldsymbol{\theta}_{qM}) \in \mathbb{R}^M$ is the vector of group-specific exposure indices for subject i, and similarly for z_j^*. The parameter η_q controls the overall variance (amplitude) of the kernel, while ρ_{qm} governs the relevance (smoothness) of the m-th index within kernel q. Larger values of ρ_{qm} result in stronger penalization for differences in index m, making the function more sensitive to variation along that dimension. This formulation allows the model to flexibly learn smooth, non-linear surfaces over the M-dimensional index space without requiring pre-specification of interactions or functional forms. The use of separate kernel matrices K_1 and K_2 enables independent modeling of baseline and trajectory effects, further increasing flexibility.

To induce sparsity and identify influential chemical exposures within each index, we assign continuous shrinkage priors to the index weights $\boldsymbol{\theta}_1$ and $\boldsymbol{\theta}_2$ [8]:

$$\theta_{qp}|\lambda_{qp}, \tau_q \sim N(0, \lambda_{qp}^2\tau_q) \tag{6}$$

where τ_q is the global parameter that shrinks all θ_{qp} toward zero; λ_{qp} is the local parameter that allows some θ_{qp} to be included in the model. The level of shrinkage is controlled by τ_q as a large value of τ_q will reflect little shrinkage and a value close to zero will shrink all θ_{qp} to zero. We also use hyperparameters as follows:

$$\lambda_{qp} \sim Cauchy(0, 1) \tag{7}$$

$$\tau_q \sim Cauchy(0, 1) \tag{8}$$

We note therefore that the horseshoe prior is a global–local shrinkage prior that allows for aggressive shrinkage of irrelevant coefficients while preserving large signals. Specifically, it combines a global shrinkage parameter that controls overall sparsity with local parameters that allow certain coefficients to "escape" shrinkage if supported by the data. This property makes it particularly well

suited for high-dimensional mixture settings, where most exposures may have negligible effects. Compared to alternatives like the spike-and-slab prior, which requires binary inclusion indicators, the continuous prior offers a computationally efficient and continuous approach to regularization that adapts to varying signal strengths. This results in more stable and interpretable estimation of index weights, facilitating identification of key chemicals within each exposure group.

When prior knowledge about the direction of effects for one or more groups is available, directional constraints can be imposed on $\boldsymbol{\theta}_{qm}$ to improve estimation.

$$\boldsymbol{b}|\sigma^2, \boldsymbol{D} \sim N_{2n}(\mathbf{0}, \sigma^2(\boldsymbol{I}_n \bigotimes \boldsymbol{D})) \tag{9}$$

where covariance matrix D follows:

$$\boldsymbol{D}^{-1} \sim Wishart_q(v_0, C_0) \tag{10}$$

$$\sigma^2 \sim gamma^{-1}(\alpha, \gamma) \tag{11}$$

$$\boldsymbol{\beta} \propto 1 \tag{12}$$

The posterior distributions were estimated using the Hamiltonian Monte Carlo (HMC) algorithm, a gradient-based Markov chain Monte Carlo method that uses simulated Hamiltonian dynamics to explore the posterior distribution more efficiently. HMC improves sampling efficiency by reducing random walk behavior and autocorrelation between samples, leading to faster convergence and more stable parameter estimates [2].

6 Simulation Study

To assess the performance of BVCMIM under various mixture effect scenarios, we designed two simulation settings that reflect common challenges in environmental health studies. Scenario A assumes all exposures with effects have the same positive direction (We chose to use positive effects in this manuscript for simplicity; however, negative effects can also be applied, they are equivalent in magnitude, the only difference is the effect direction during interpretation, and thus not necessary to include.) of contributions, whereas Scenario B includes both protective (positive) and harmful (negative) chemicals, introducing mixed (bidirectional) effects. In each scenario, a subset of exposure groups was assigned true effects on the baseline outcome and/or the outcome trajectory. We also introduced known interactions between groups to evaluate BVCMIM's ability to recover non-additive relationships. Table 4 summarizes the configuration of each simulation scenario.

More specifically, in our simulated examples, we first wanted to mimic the empirical correlation structure of real exposures from the HOME dataset. In order to do that, we calculated the empirical covariance matrix $\boldsymbol{\Sigma}$ derived from 36 chemical exposures grouped into $M = 5$ mutually exclusive categories. For

each simulated subject i, we then generated a $P = 36$-dimensional exposure vector from a multivariate normal distribution:

$$\boldsymbol{z}_i \sim \text{MVN}_{36}(\mathbf{0}, \boldsymbol{\Sigma}).$$

We assigned group-specific index weights $\boldsymbol{\theta}_{qm}$ that differ for baseline ($q = 1$) and trajectory ($q = 2$) effects. The exposure indices for each subject were then computed as:

$$\boldsymbol{z}_i^* = \left(\boldsymbol{z}_{i1}^T\boldsymbol{\theta}_{q1}, \ldots, \boldsymbol{z}_{iM}^T\boldsymbol{\theta}_{qM}\right),$$

which are passed through index-response functions $h_q(\cdot)$, where $q = 1, 2$. We simulated two distinct scenarios:

- **Scenario A (Positive Effects Only):** In this setting, all non-zero weights are positive. The assigned weights are:
  $\boldsymbol{\theta}_{11} = (0.5, 0.6, 0.5, 0, 0, 0, 0, 0)^T$, $\boldsymbol{\theta}_{15} = (0, 0, 0.5, 0.5)^T$,
  $\boldsymbol{\theta}_{22} = (0.6, 0.4, 0.5, 0, 0, 0, \ldots, 0)^T$, $\boldsymbol{\theta}_{24} = (0, 0, 0, 0, 0.4, 0.2)^T$. All other weights are set to zero. The index-response functions are:

  $$h_{i1} = \boldsymbol{z}_{i1}^T\boldsymbol{\theta}_{11} + f_A\left(\boldsymbol{z}_{i5}^T\boldsymbol{\theta}_{15}\right), \quad h_{i2} = \boldsymbol{z}_{i2}^T\boldsymbol{\theta}_{22} + f_A\left(\boldsymbol{z}_{i4}^T\boldsymbol{\theta}_{24}\right),$$

  where $f_A(\cdot)$ is an S-shaped (sigmoidal) function capturing non-linear effects.

- **Scenario B (Mixed Positive and Negative Effects):** This scenario includes both protective and harmful exposures. The assigned weights are: $\boldsymbol{\theta}_{11} = (0.5, 0.6, 0.5, 0, 0, 0, 0, 0)^T$, $\boldsymbol{\theta}_{13} = (-0.8, -0.8)^T$,
  $\boldsymbol{\theta}_{15} = (0, 0, 0.5, 0.5)^T$, $\boldsymbol{\theta}_{22} = (0.6, 0.4, 0.5, 0, 0, 0, \ldots, 0)^T$,
  $\boldsymbol{\theta}_{23} = (-0.8, -0.8)^T$, $\boldsymbol{\theta}_{24} = (0, 0, 0, 0, 0.4, 0.2)^T$. All other weights are zero. The index-response functions are:

  $$h_{i1} = \boldsymbol{z}_{i1}^T\boldsymbol{\theta}_{11} + \boldsymbol{z}_{i3}^T\boldsymbol{\theta}_{13} + f_B\left(\boldsymbol{z}_{i5}^T\boldsymbol{\theta}_{15}\right), \quad h_{i2} = \boldsymbol{z}_{i2}^T\boldsymbol{\theta}_{22} + \boldsymbol{z}_{i3}^T\boldsymbol{\theta}_{23} + f_B\left(\boldsymbol{z}_{i4}^T\boldsymbol{\theta}_{24}\right),$$

  where $f_B(\cdot)$ is again an S-shaped function capturing non-linear effects.

Table 5 presents the results of simulations. In each scenario, we generated 10 datasets of 100 observations. We assessed the performance of estimating effects $\boldsymbol{h}$ by regressing the predicted $\hat{h}$ on grand truth $\boldsymbol{h}$ for h_1 and h_2. We present the intercept, slope, R^2 and root mean squared error (RMSE) of the regressions over 10 datasets. Good performance on estimation can be indicated when the intercept is close to zero, and the slope and R^2 are both close to 1. We see that the performance is consistent across all simulations; for all simulations, the intercept ranging from -0.05 to 0.176, the slope ranges from 0.950 to 1.001, and the R^2 ranges from 0.972 to 0.990. The result shows adding directional constrains from prior knowledge doesn't improve the performance on estimating $\boldsymbol{h}$. Lastly, the RMSE is higher for h_1 than h_2.

We further identify the indices of the mixtures most associated with health outcomes. Drawing inspiration from relative importance, which has been defined in previous literature to quantify the impact of each chemical on health

outcomes [20], we propose an index-wise relative importance metric to quantify the impact of each chemical index on both the baseline health outcome and the outcome trajectory. This metric is based on predicting the baseline outcome (h_1) and the outcome trajectory (h_2) using index-response functions at two new exposure profiles: one where all chemicals in an index group are at high exposure (75th percentile) and another where they are at low exposure (25th percentile).For example, to estimate the relative importance of chemical index z_1^*, we predict the index-response functions h_1 and h_2 for high and low z_1^*, while holding chemicals in other indices of the mixture at their median exposure to isolate the effect of z_1^*. The difference in the estimated h_1 for high versus low z_1^* corresponds to the relative importance of index z_1^* on the baseline outcome. Similarly, the difference in the estimated h_2 for high versus low z_1^* corresponds to the relative importance of index z_1^* on the outcome trajectory.

The relative importance for scenario A and B are given in figure 1 and 2, in which we plot the BVCMIM predicted versus true relative importance for each simulation. The result for each simulation scenario were averaged across 10 datasets. For each simulation, the relative importance was estimated on the dataset for that simulation, whereas the true known relative importance remains the same across datasets. 95% confidence interval are given as the predicted relative importance $\pm$ 1.96 times the square root of the predicted corresponding variance. We see that, in both scenarios, the most 95% intervals capture the true index-response effect. For a few intervals which did not capture the index-response effect exactly, they are very close to the true effects. Relative importance plots show that chemical indices which are most associated with health baseline outcome and outcome trajectory are welly estimated by BVCMIM.

A feature of BVCMIM is that it is simple and straightforward to interpret interaction as we only need to investigate $M \times (M-1)$ index wise interactions. We present two-way index wise interactions for both baseline outcome and outcome trajectory for scenario A and scenario B in figure 3 and 4 . The interaction between index i and index j are visualized as the index-response curves of index i while fix index j at low (25th percentile) and high (75th percentile) and all other indices at their median. Parallel lines indicates no interaction between indices while deviation indicates interactions. We can see from figure 3 and figure 4 that, at baseline outcome, there is a strong interaction between index 1 and index 5, and at outcome trajectory, there is a strong interaction between index 2 and index 4. They are consistent with simulation generating processes.

BVCMIM also provides estimation of weights for each chemical exposure which can be interpreted as contributions to the indices which they belong to. Table 6 shows the estimated weights from simulation scenario A. At baseline, z_1, z_2 and z_3 are significant in index 1, $z35$ and $z36$ are significant in index 5; at the outcome trajectory, z_9, z_{10} and z_{11} are significant in index 2, z_{31} and z_{32} are significant in index 4. These estimations match with the setting

in simulation scenario A. Similarly, table 7 shows the estimated weights from simulation scenario B. At baseline, z_1, z_2 and z_3 are significant in index 1, z_{25} and z_{26} are significant in index 3 and z_{35} and z_{36} are significant in index 5. At the outcome trajectory, z_9, z_{10} and z_{11} are significant in index 2, z_{25} and z_{26} are significant in index 3 and z_{31} and z_{32} are significant in index 4. These estimations match with the setting in simulation scenario B.

7 Application to the HOME Study

To illustrate the practical utility of BVCMIM, we applied the model to data from the Health Outcomes and Measures of the Environment (HOME) Study—a prospective birth cohort designed to investigate environmental influences on child development. This study provides rich longitudinal outcome data and detailed prenatal chemical exposure measurements, making it an ideal setting for evaluating the temporal dynamics of exposure-response relationships.

Prior research using HOME data has shown that several environmental chemicals, particularly endocrine-disrupting compounds such as phthalates, bisphenol A (BPA), and perfluorinated compounds (PFCs), are associated with altered neurodevelopmental outcomes in early childhood [4, 16]. However, most existing analyses have relied on single-chemical models or linear mixture approaches, which may fail to capture the complex, non-linear, and time-varying effects of chemical mixtures. Moreover, the presence of correlated exposures and potential interactions between chemical groups presents an additional layer of analytical complexity.

BVCMIM addresses these challenges by enabling simultaneous estimation of baseline effects and outcome trajectories, while also accounting for non-linearity, interactions between exposure groups, and individual chemical contributions. In our analysis, Age was treated as a discrete time variable, with 12 months coded as baseline (0), 24 months as 1, and 36 months as 2. A Gaussian kernel was used to flexibly model the index-response surface. The Markov Chain Monte Carlo (MCMC) sampler was run for 2,000 iterations, discarding the first 1,000 as burn-in.

Each exposure index corresponds to a distinct chemical group. Figure 5 shows the estimated relative importance of each index at baseline and over the outcome trajectory. At baseline, indices corresponding to phthalates & BPA and perfluorinated compounds (PFCs) were negatively and positively associated with MDI scores, respectively.

To examine potential synergistic effects, we visualized the interaction between the phthalates & BPA group and the PFC group in Figure 6. A modest interaction effect was observed, suggesting potential modification of one group's effect depending on the level of the other.

Table 8 reports the estimated weights for individual chemicals within each group. Among phthalates & BPA, mono-(3-carboxypropyl) phthalate (MCPP) had the strongest influence on neurodevelopmental outcomes. Within the PFC group, perfluorononanoic acid (PFNA) was identified as a key contributor, show-

ing a significant association with MDI scores.

8 Discussion

We introduced the Bayesian Varying Coefficient Multiple Index Model (BVCMIM), a novel hierarchical modeling framework that integrates mixed-effects and multiple-index modeling to evaluate the longitudinal effects of environmental chemical mixtures on health outcomes. BVCMIM is specifically designed to estimate both baseline effects and outcome trajectories, addressing the complexity and high dimensionality of chemical exposure data.

A key feature of BVCMIM is the construction of group-based exposure indices, which reduce dimensionality while allowing for flexible, non-linear index-response relationships. By applying the horseshoe prior, the model enables automatic shrinkage and selection of influential chemicals within each group. Furthermore, by incorporating interaction terms between indices, BVCMIM allows for the exploration of synergistic or antagonistic effects across chemical groups.

Posterior inference was performed using Hamiltonian Monte Carlo (HMC), which offers efficient sampling in high-dimensional parameter spaces. Our simulation studies demonstrated that BVCMIM accurately estimates both main effects and interactions under various exposure scenarios, including those with unidirectional and bidirectional effects. Notably, adding directional constraints based on prior knowledge did not substantially improve performance, suggesting that the horseshoe prior sufficiently manages sparsity in most settings.

When applied to the HOME Study cohort, BVCMIM identified meaningful associations between prenatal exposure indices and neurodevelopmental outcomes. At baseline, Phthalates & BPA were negatively associated with children's neurodevelopment, whereas PFCs showed a positive association. Similar findings can be found in previous publications for the inverse association between Phthalates & BPA with children's neurodevelopment [19]. The model also detected a modest interaction effect between these two groups. Chemical-specific weights highlighted MCPP and PFNA as key contributors within their respective indices, aligning with existing toxicological literature [29, 23, 26, 17].

While BVCMIM provides a powerful and interpretable framework, it has several limitations. First, chemical exposures were measured only once during pregnancy, and the model does not currently accommodate time-varying or postnatal exposures, which may influence developmental outcomes. Second, exposure groupings were predefined based on chemical class or structure, and any misclassification in these groupings could affect interpretability and estimation. Future extensions could incorporate data-driven or hierarchical grouping strategies. Third, the Gaussian kernel was used to model non-linear index-response surfaces, but performance may vary depending on kernel choice and tuning parameters; sensitivity analyses were not conducted in this work. Additionally, while the horseshoe prior performs well for sparsity, sensitivity to hyperparameter choices was not fully explored in this study. More genarally, we also note

that the HOME dataset includes a relatively modest sample size (n = 149), which may limit power to detect small or complex interaction effects. Finally, as with any observational study, unmeasured confounding cannot be ruled out, and causal interpretations should be made cautiously.

In summary, BVCMIM offers a flexible and scalable approach for analyzing complex chemical mixtures in longitudinal studies. It balances interpretability, modeling flexibility, and computational efficiency—allowing researchers to uncover both main effects and interactions among chemical groups. By identifying key contributors within mixtures and capturing how their effects evolve over time, BVCMIM provides valuable insights for environmental health research and risk assessment. Future extensions incorporating time-varying exposures and alternative grouping structures will further broaden its utility.

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Table 1: Demographic Profile of Participants

Category	**Subgroup**	**N**	**Percentage (%)**
Overall		149	
Child gender	Female	86	57.7
	Male	63	42.3
Maternal age	<25	25	16.8
	25−34	98	65.8
	≥35	26	17.4
Maternal race	White	108	72.5
	Black	34	22.8
	Other	7	4.6
Maternal education	Graduate school	34	22.8
	College	93	62.4
	High school	19	12.8
	< High school	6	4.0
Marital status	Married	114	76.5
	Not Married	35	23.5
Annual household income	≥ $8000	41	27.5
	$40,000 to <$80,000	63	42.3
	$20,000 to <$40,000	21	14.1
	<$20,000	24	16.1
Maternal IQ	1st tertilt (58-101)	40	26.8
	2nd tertile (>101-114)	59	39.6
	3rd tertile (115-134)	40	26.8
Maternal insurance	Private	121	81.2
	Public/None	28	18.8

Table 2: Geometric means and standard deviations of maternal biological samples chemicals concentrations during pregnancy for all 149 participants

Category	**Chemical**	**GM (GSD)**
Pth & BPA	MBP	22.6 (1.3)
	MIBP	5.7 (1.0)
	MEP2	95.6 (1.5)
	MBZP2	9.4 (1.2)
	MCPP	3.0 (0.7)
	MEHHP	24.65 (1.35)
	MECPP	35.28 (1.33)
	BPA	3.13 (0.87)
PCB	PCB28	8.17 (0.54)
	PCB74	18.46 (0.52)
	PCB99	18.28 (0.56)
	PCB105	8.25 (0.58)
	PCB118	30.77 (0.62)
	PCB138_158	50.01 (0.63)
	PCB146	8.20 (0.59)
	PCB153	69.50 (0.64)
	PCB156	11.78 (0.64)
	PCB170	18.88 (0.70)
	PCB183	7.19 (0.58)
	PCB187	14.49 (0.64)
	PCB194	9.83 (0.64)
	PCB196_203	11.64 (0.61)
	PCB199	9.82 (0.65)
	PCB206	6.59 (0.50)
OC	PP_DDE	452.74 (0.61)
	T_NONA	48.40 (0.62)
PBDE	BB153	8.33 (0.69)
	PBDE28	8.15 (0.77)
	PBDE99	29.93 (1.07)
	PBDE100	26.10 (0.98)
	PBDE153	34.43 (1.01)
PFC	PFOA	6.80 (0.45)
	PFOS	14.28 (0.45)
	PFNA	2.03 (0.20)
	PFHXS	2.73 (0.43)

Table 3: Means and Standard Deviations of MDI Score at Each Age

Age (Month)	**Mean (SD)**
12	94.01 (9.59)
24	90.99 (12.86)
36	94.11 (12.49)

Table 4: Simulation scenario design summary for evaluating BVCMIM.

Scenario A: Positive Effects Only	
Effect direction	All effects are positive
Groups with baseline effects	Group 1, Group 5
Groups with trajectory effects	Group 2, Group 4
Interaction structure	Group 1 × Group 5 (baseline) Group 2 × Group 4 (trajectory)
Scenario B: Mixed Positive and Negative Effects	
Effect direction	Mixed: both positive and negative effects
Groups with baseline effects	Group 1 (positive), Group 3 (negative), Group 5 (positive)
Groups with trajectory effects	Group 2 (positive), Group 3 (negative), Group 4 (positive)
Interaction structure	Group 1 × Group 5 (baseline) Group 2 × Group 4 (trajectory)

Table 5: Simulation results evaluating the estimation accuracy of BVCMIM under Scenarios A and B. Each row summarizes the regression of predicted $\hat{h}$ on true h across 10 simulated datasets. Metrics include intercept, slope, R^2, and root mean squared error (RMSE).

Scenario	**Constrain**	**h**	**Intercept**	**Slope**	R^2	**RMSE**
A	No	1	0.05 (0.03)	0.95 (0.01)	0.97 (0.01)	0.48 (0.03)
		2	0.06 (0.03)	1.00 (0.01)	0.99 (0.01)	0.33 (0.05)
A	Yes	1	0.06 (0.03)	0.97 (0.01)	0.97 (0.01)	0.47 (0.04)
		2	0.03 (0.07))	1.00 (0.01)	0.98 (0.01))	0.42 (0.05)
B	No	1	0.08 (0.03)	0.95 (0.02)	0.97 (0.01)	0.53 (0.04)
		2	0.01 (0.03)	1.00 (0.01)	0.99 (0.01)	0.28 (0.07)
B	Yes	1	0.18 (0.03)	0.95 (0.01)	0.98 (0.01)	0.49 (0.02)
		2	-0.05 (0.04)	0.98 (0.02)	0.97 (0.01)	0.48 (0.01)

Table 6: Estimated posterior weights and 95% credible intervals for each chemical in Simulation Scenario A.

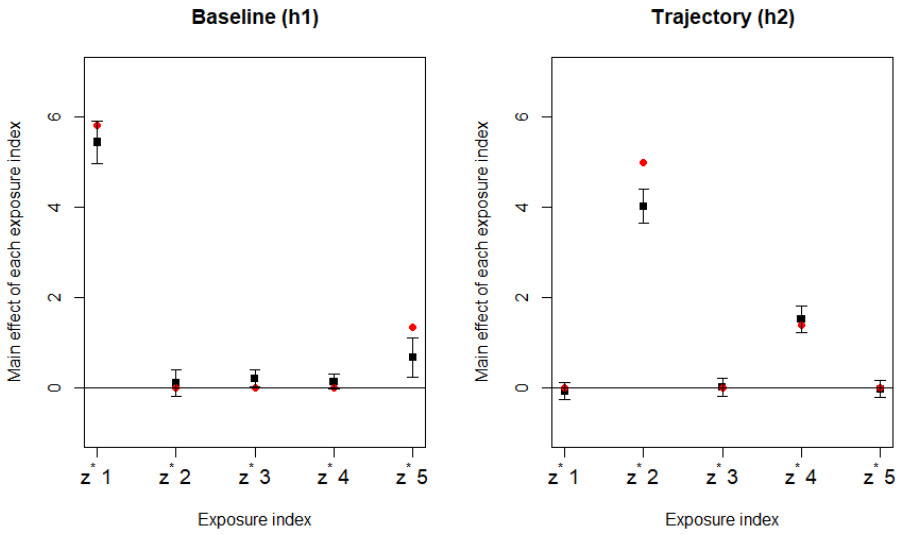
h	Group	Exposure	Weights	95% CI
h_1	z_1^*	$z1$	13.99	(0.16, 190.19)
		$z2$	16.15	(0.17, 216.44)
		$z3$	14.51	(0.15, 191.20)
		$z4$	0.06	(-0.99, 1.02)
		$z5$	-0.78	(-13.45, 0.28)
		$z6$	0.20	(-1.15, 2.90)
		$z7$	-0.09	(-1.75, 1.75)
		$z8$	-1.95	(-25.32, 0.01)
	z_5^*	$z33$	1.60	(-0.01, 15.74)
		$z34$	0.61	(-0.14, 8.24)
		$z35$	12.24	(0.10, 117.19)
		$z36$	4.92	(0.04, 47.50)
h_2	z_2^*	$z9$	22.85	(0.25, 219.21)
		$z10$	18.03	(0.19, 178.38)
		$z11$	24.22	(0.27, 227.46)
		$z12$	-0.31	(-4.09, 1.79)
		$z13$	-1.01	(-7.36, 0.44)
		$z14$	-0.13	(-4.55, 3.40)
		$z15$	-0.67	(-13.12, 3.46)
		$z16$	-0.22	(-6.09, 2.25)
		$z17$	-0.16	(-7.87, 2.36)
		$z18$	0.11	(-5.85, 7.50)
		$z19$	-0.30	(-4.86, 1.67)
		$z20$	0.35	(-3.24, 9.25)
		$z21$	-1.26	(-14.57, 0.82)
		$z22$	-0.88	(-8.51, 2.10)
		$z23$	3.32	(-0.35, 34.36)
		$z24$	-1.54	(-25.64, 2.92)
	z_4^*	$z27$	-1.35	(-12.53, 0.29)
		$z28$	-3.47	(-46.81, 2.99)
		$z29$	-0.28	(-13.27, 6.44)
		$z30$	-0.08	(-3.46, 3.75)
		$z31$	34.93	(0.37, 340.19)
		$z32$	13.76	(0.12, 137.70)

Table 7: Estimated posterior weights and 95% credible intervals for each chemical in Simulation Scenario B.

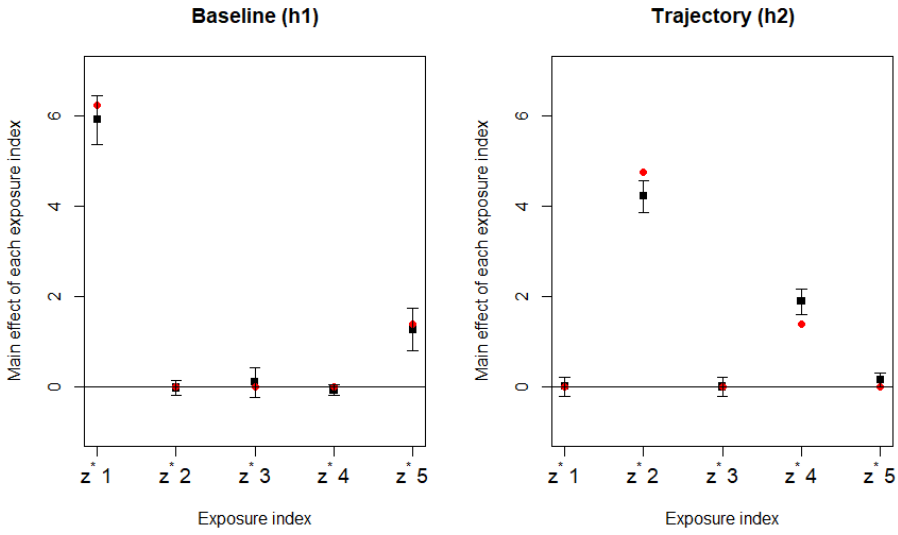
h	Group	Exposure	Weights	95% CI
h_1	z_1^*	$z1$	-7.60	(-45.87, -0.24)
		$z2$	-8.52	(-50.49, -0.26)
		$z3$	-8.31	(-53.59, -0.24)
		$z4$	-0.02	(-20.99, 0.14)
		$z5$	0.64	(-0.01, 5.36)
		$z6$	-0.14	(-0.79, 0.01)
		$z7$	-0.11	(-2.22, 0.49)
		$z8$	-0.13	(-1.47, 0.42)
	z_3^*	z_{25}	1.06	(0.03, 9.74)
		z_{26}	0.61	(0.02, 5.46)
	z_5^*	$z33$	-0.38	(-1.71, 0)
		$z34$	-0.08	(-0.46, 0.04)
		$z35$	1.61	(0.08, 7.99)
		$z36$	1.54	(0.07, 7.76)
h_2	z_2^*	$z9$	5.50	(0.05, 36.49)
		$z10$	4.41	(0.05, 28.46)
		$z11$	4.29	(0.01, 29.59)
		$z12$	0.17	(-0.27, 1.18)
		$z13$	-0.30	(-2.98, 0.31)
		$z14$	0.68	(0.01, 3.38)
		$z15$	-0.19	(-0.80, -0.01)
		$z16$	-1.03	(-6.51, 0.42)
		$z17$	0.04	(-0.37, 0.85)
		$z18$	0	(-0.82, 0.70)
		$z19$	-0.62	(-4.10, 0.13)
		$z20$	-0.15	(-1.41, 0.45)
		$z21$	3.02	(0.03, 17.51)
		$z22$	-0.91	(-5.58, -0.02)
		$z23$	-0.61	(-4.30, 0.07)
		$z24$	-0.29	(-2.23, 0.04)
	z_3^*	z_{25}	0.81	(0.01, 5.04)
		z_{26}	1.00	(0.01, 6.98)
	z_4^*	$z27$	-0.18	(-2.31, 0.56)
		$z28$	1.01	(-0.47, 7.25)
		$z29$	1.12	(-0.98, 9.29)
		$z30$	2.88	(0.06, 14.43)
		$z31$	-22.74	(-83.54, -1.18)
		$z32$	-8.33	(-32.51, -0.37)

Table 8: Posterior estimates of chemical-specific weights at baseline for HOME Cohort, with 95% credible intervals

Group	**Exposure**	**Weights**	**95% CI**
Pth & BPA	MBP	1.01	(-2.52, 10.92)
	MIBP	0.68	(-4.10, 9.64)
	MEP2	1.09	(-3.20, 10.90)
	MBZP2	2.62	(-3.52, 10.86)
	MCPP	60.90	(0.60, 323.18)
	MEHHP	1.45	(-1.78, 13.80)
	MECPP	-1.06	(-13.82, 3.88)
	BPA	10.86	(-2.10, 78.42)
PCB	PCB28	-1.30	(-12.94, 6.45)
	PCB74	2.02	(-7.76, 25.79)
	PCB99	2.51	(-16.26, 14.25)
	PCB105	5.18	(-4.51, 34.60)
	PCB118	2.91	(-8.28, 26.47)
	PCB138_158	4.09	(-7.14, 51.30)
	PCB146	1.84	(-8.33, 13.55)
	PCB153	1.70	(-18.58, 18.35)
	PCB156	64.05	(-0.37, 603.37)
	PCB170	46.90	(-16.90, 339.75)
	PCB183	33.56	(-4.55, 177.45)
	PCB187	27.00	(-3.79, 55.98)
	PCB194	0.85	(-8.04, 11.80)
	PCB196_203	10.10	(-28.86, 77.21)
	PCB199	4.38	(-5.55, 16.58)
	PCB206	10.24	(-19.98, 87.44)
OC	PP_DDE	-0.41	(-8.40, 1.99)
	T_NONA	-43.25	(-449.76, -0.13)
PBDE	BB153	34.83	(-1.68, 133.41)
	PBDE28	0.15	(-2.62, 15.80)
	PBDE47	0.40	(-6.04, 12.53)
	PBDE99	11.42	(-4.37, 75.44)
	PBDE100	4.66	(-3.86, 13.61)
	PBDE153	0.12	(-4.94, 4.10)
PFC	PFOA	-1.15	(-7.66, 2.38)
	PFOS	-3.07	(-14.78, 2.83)
	PFNA	-12.21	(-92.54, -0.02)
	PFHXS	1.59	(-3.51, 15.02)

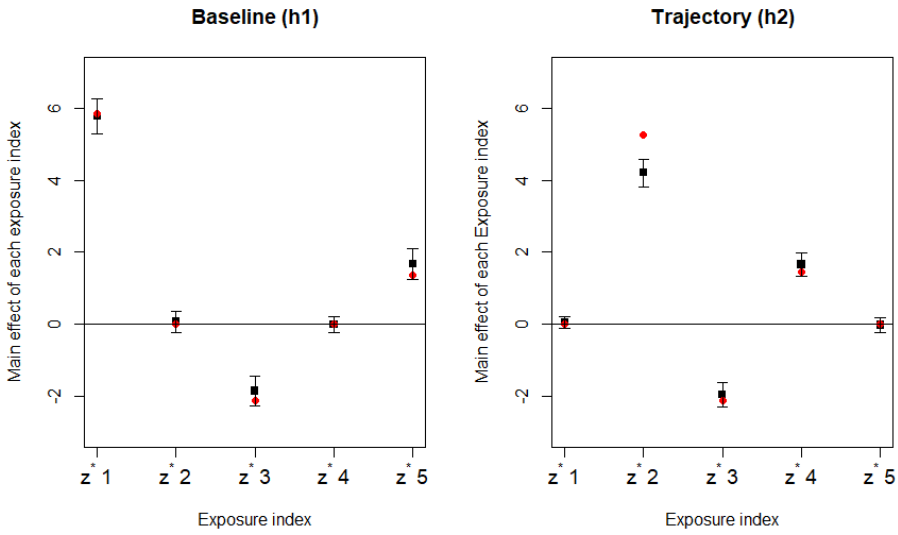


(a) Without directional constrain

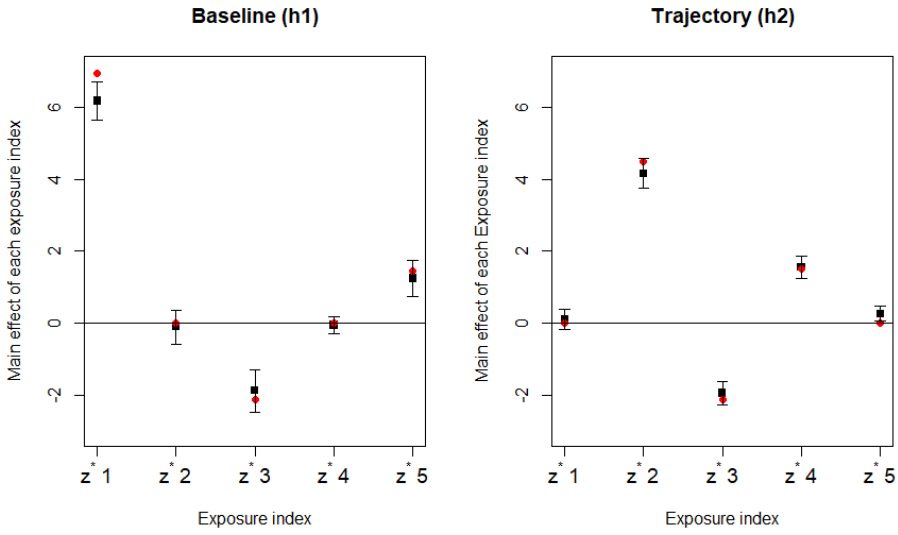


(b) With directional constrain

Figure 1: Comparison of BVCMIM-predicted versus true relative importance for chemical indices in Simulation Scenario A (unidirectional effects). (a) Without directional constraints; (b) With directional constraints.

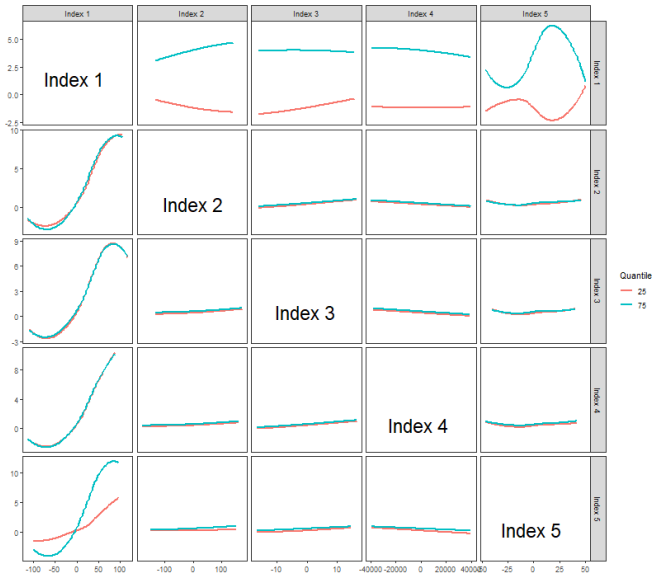


(a) Without directional constrain

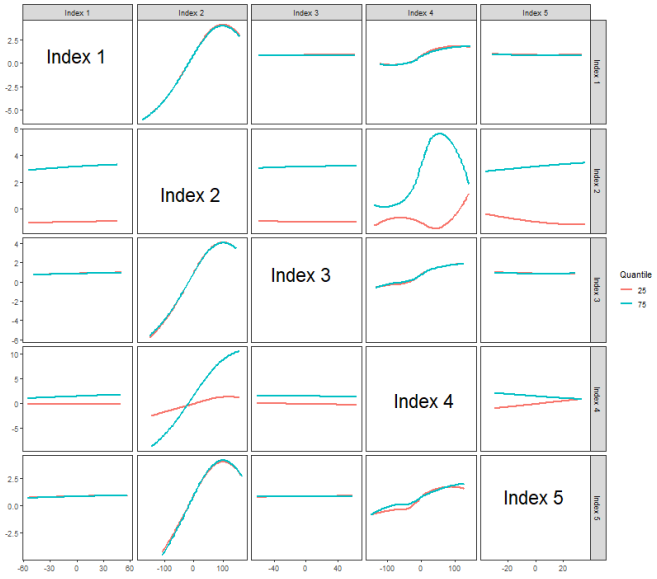


(b) With directional constrain

Figure 2: Comparison of BVCMIM-predicted versus true relative importance for chemical indices in Simulation Scenario B (bidirectional effects). (a) Without directional constraints; (b) With directional constraints.

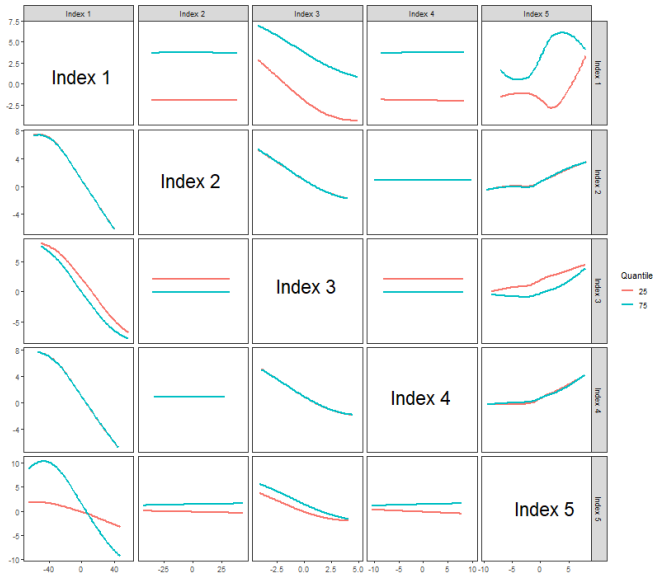


(a) Interaction plot at baseline (h_1)

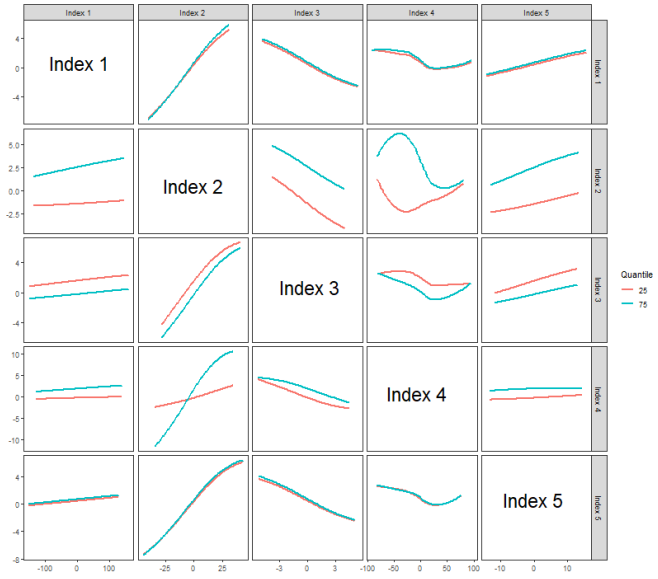


(b) Interaction plot at Tragectory (h_2)

Figure 3: Visualizing two-way interaction of scenario A with BVCMIM. Each panel shows the index-response relation for one index with all other indices fixed at a given percentile. Panel (a) is for h_1 and panel (b) is for h_2

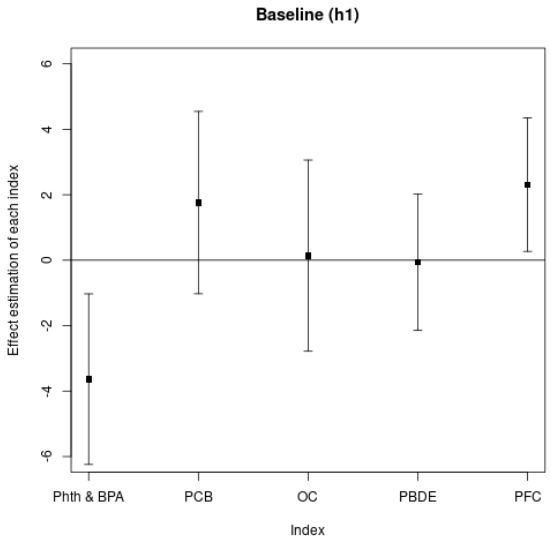


(a) Interaction plot at Baseline (h_1)

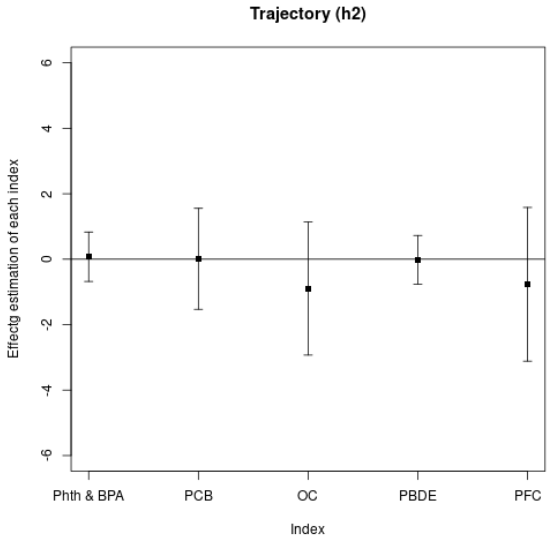


(b) Interaction plot at Tragectory (h_2)

Figure 4: Visualizing two-way interaction of scenario B with BVCMIM. Each panel shows the index-response relation for one index with all other indices fixed at a given percentile. Panel (a) is for h_1 and panel (b) is for h_2



(a) Relative importance at baseline



(b) Relative importance at trajectory

Figure 5: Estimated relative importance of chemical indices in the HOME cohort. (a) Baseline importance at 12 months; (b) Importance over the outcome trajectory from 12 to 36 months.

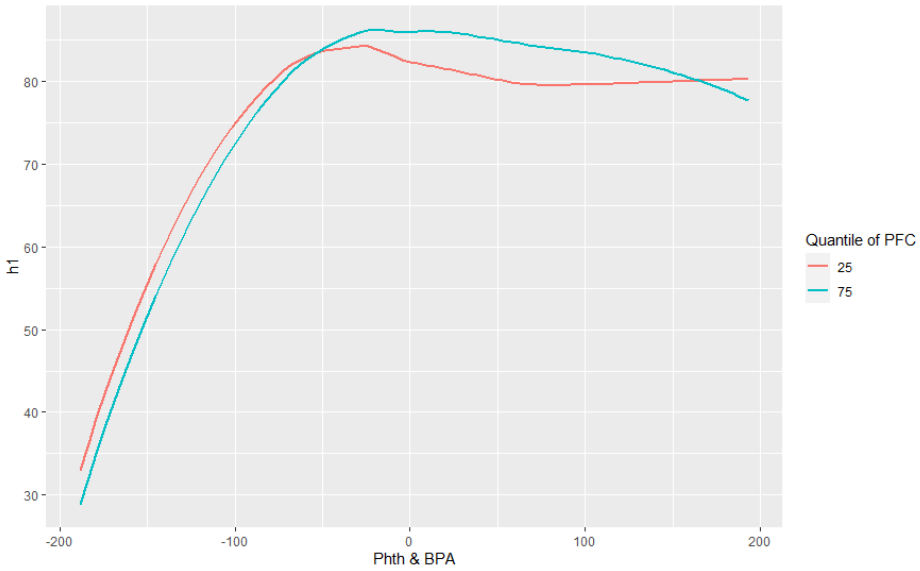


Figure 6: Estimated interaction effect between the Phthalates & BPA group and the PFC group on children's neurodevelopment.